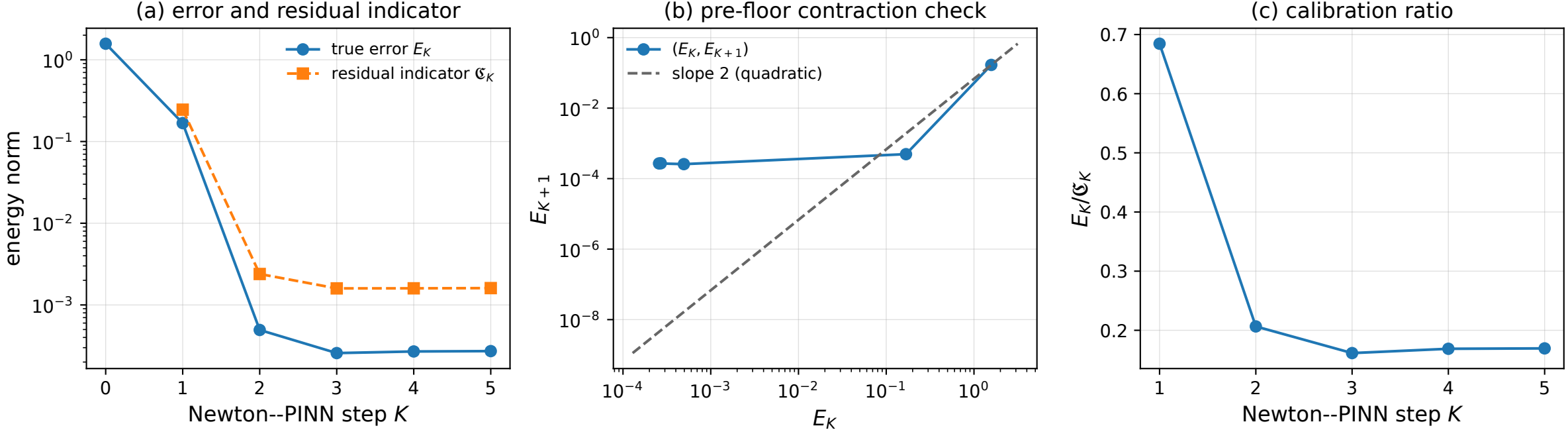

(a) error and residual indicator
true error E_K
residual indicator $\mathfrak{C}_K$
energy norm
Newton--PINN step K
(b) pre-floor contraction check
(E_K, E_{K+1})
slope 2 (quadratic)
E_{K+1}
E_K
(c) calibration ratio
$E_K/\mathfrak{C}_K$
Newton--PINN step K